National Air Quality Analysis and Prediction with Weather Impact (India)

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Abstract

Air pollution remains one of the most critical public health challenges in urban India. This paper presents a comprehensive analysis of air quality data collected from four major Indian cities — Delhi, Mumbai, Kolkata and Bengaluru — over a three-year period from 2022 to 2024. Daily data sourced from the Central Pollution Control Board (CPCB) was preprocessed, cleaned and analyzed to identify pollution trends, seasonal patterns and the meteorological influence on PM2.5 levels, used here as a proxy for the Air Quality Index (AQI). Exploratory data analysis revealed significant inter-city variation with Delhi recording the highest average PM2.5 at 127.71 µg/m3 and Bengaluru the lowest at 28.60 µg/m3. Strong seasonal patterns were identified with pollution peaking in winter months (November to January) and declining during the monsoon season (June to August). Negative correlations were observed between PM2.5 and temperature (r = -0.406) and between PM2.5 and wind speed (r = -0.095). Three machine learning models — Multiple Linear Regression, Random Forest Regression and K-Nearest Neighbors (KNN) — were trained and compared for PM2.5 prediction. Random Forest achieved the best predictive accuracy across all evaluation metrics.

Keywords — *Air Quality Index, PM2.5, CPCB, Exploratory Data Analysis, Random Forest, KNN, Linear Regression, Seasonal Analysis, India*

I. INTRODUCTION

Air pollution poses a severe threat to public health across Indian metropolitan areas. Rapid industrialization, rising vehicular density and urban expansion have significantly degraded ambient air quality. The World Health Organization (WHO) annual safe limit for PM2.5 is 15 µg/m3 [1]. Indian cities consistently exceed this benchmark, with Delhi frequently ranked among the most polluted cities globally.

The Central Pollution Control Board (CPCB) operates a network of Continuous Ambient Air Quality Monitoring Stations (CAAQMS) across India. This study uses daily data from four CPCB stations — Anand Vihar (Delhi), Colaba (Mumbai), Rabindra Bharati (Kolkata) and BTM Layout (Bengaluru) — spanning January 2022 to December 2024.

This study addresses three objectives: (1) to analyze trends and patterns of major air pollutants including PM2.5, PM10, NO2, SO2, CO and O3 across four cities over three years; (2) to examine the relationship between meteorological factors including temperature, humidity, wind speed and rainfall with air quality; and (3) to develop and compare machine learning models for forecasting PM2.5 as a proxy for AQI.

II. RELATED WORK

Kumar and Pande investigated six years of air pollution data from 23 Indian cities sourced from CPCB for air quality analysis and AQI prediction. Their study applied five machine learning models after correlation-based feature selection and exploratory data analysis. The XGBoost model achieved the best performance while Gaussian Naive Bayes achieved the highest classification accuracy. Their work also identified a significant drop in pollutant levels during 2020 due to COVID-19 lockdowns.

Doreswamy applied multiple machine learning regression models to forecast PM2.5 concentrations using Indian city data. Their study confirmed that PM2.5 and PM10 exhibit clear seasonal variation with higher concentrations observed in winter months and significantly lower values during the monsoon season. Linear Regression, Decision Tree and Random Forest were among the models evaluated in their work.

Breiman introduced the Random Forest algorithm as an ensemble of decision trees using bagging to reduce variance and improve generalization. The method has since been widely adopted in environmental prediction tasks including air quality forecasting due to its ability to capture non-linear relationships between input features and target variables without requiring feature scaling.

Pedregosa presented scikit-learn, an open-source machine learning library for Python that provides implementations of Linear Regression, Random Forest, KNN and other algorithms used in this study. Their framework has become the standard tool for machine learning experiments in environmental and data science research.

III. DATASET AND PREPROCESSING

A. Data Source

Daily air quality data was downloaded from the CPCB open data portal at airquality.cpcb.gov.in for one monitoring station per city across three calendar years (2022, 2023 and 2024), producing 12 individual CSV files with a total of 1096 daily records per city after merging.

B. Features

Each record contains 25 features including pollutant concentrations: PM2.5, PM10, NO, NO2, NOx, NH3, SO2, CO, Ozone and Benzene; and meteorological parameters:

ambient temperature (AT), relative humidity (RH), wind speed (WS), wind direction (WD), total rainfall (TOT-RF), solar radiation (SR) and barometric pressure (BP).

C. Preprocessing

Three yearly CSV files per city were concatenated using pandas and sorted by date index. Missing values were handled using forward fill followed by backward fill. This preserves temporal continuity without introducing artificial spikes as mean or median imputation would in daily time series data. Columns with 100% missing values across all cities (Xylene, O Xylene) were dropped. Columns with 75% or more missing data (RF, VWS) were also dropped. All four city dataframes were merged into one superset of 4384 rows with a City column added.

TABLE I. DATASET SUMMARY

City	Station	Records	Avg PM2.5 (µg/m3)
Delhi	Anand Vihar	1096	127.71
Kolkata	Rabindra Bharati	1096	49.28
Mumbai	Colaba	1096	39.68
Bengaluru	BTM Layout	1096	28.60

IV. EXPLORATORY DATA ANALYSIS

A. City-wise Pollution Levels

Delhi recorded the highest average PM2.5 at 127.71 µg/m3, followed by Kolkata (49.28 µg/m3), Mumbai (39.68 µg/m3) and Bengaluru (28.60 µg/m3). All four cities exceeded the WHO safe limit of 15 µg/m3. Delhi's average PM10 of 305.72 µg/m3 is nearly three times higher than Kolkata (106.52 µg/m3). Delhi also had the highest NO2 (78.7 µg/m3) and SO2 (17.8 µg/m3) concentrations among the four cities.

B. Seasonal Patterns

Monthly analysis reveals a clear seasonal cycle across all cities. Average PM2.5 peaks in January (111.01 µg/m3) and November (110.19 µg/m3) due to winter temperature inversions and crop residue burning. The lowest values occur in July (23.36 µg/m3) during the monsoon season when rainfall scavenges particulates from the atmosphere. This seasonal pattern is consistent with findings reported by Doreswamy for Indian cities.

C. Meteorological Correlations

Ambient temperature (AT) showed the strongest negative correlation with PM2.5 (r = -0.406). Relative humidity showed a negative correlation (r = -0.306) and wind speed showed a negative correlation (r = -0.095). PM2.5 and PM10 were strongly positively correlated (r = 0.894) indicating shared emission sources. CO, NO2 and NOx showed moderate positive correlations with PM2.5 pointing to vehicular emissions as a major contributor.

D. Outlier Analysis

IQR-based outlier detection identified significant outliers in SO2, NH3 and Benzene due to high positive skewness. Delhi recorded the highest outlier counts in PM2.5 and PM10 corresponding to severe winter pollution episodes. These outliers were retained as they represent real atmospheric events rather than measurement errors, consistent with the approach adopted by Kumar and Pande [3].

V. MACHINE LEARNING MODELS

A. Target Variable and Features

PM2.5 was selected as the prediction target as a proxy for AQI. This decision is grounded in its dominant role in India's National AQI calculation as defined by CPCB. The feature set included 13 variables: PM10, NO2, NOx, NO, NH3, SO2, CO, Ozone, RH, WS, WD, SR and BP, covering both pollution and meteorological factors.

B. Train-Test Split

The dataset was split into 80% training (3507 samples) and 20% testing (877 samples) using random_state=42 for reproducibility, following standard machine learning practice as implemented in scikit-learn.

C. Multiple Linear Regression

Multiple Linear Regression was applied as a baseline model. It estimates PM2.5 as a linear combination of input features. While computationally efficient, it cannot capture non-linear interactions present in pollution data. It provides a performance benchmark for comparison.

D. Random Forest Regression

Random Forest Regression with 100 estimators was applied. As described by Breiman, this ensemble method reduces variance through bagging of decision trees and captures non-linear feature interactions without requiring feature scaling. Feature importance scores were extracted to identify the most influential predictors.

E. KNN Regression

KNN Regression with k=5 was implemented using scikit-learn. Since KNN is distance-based and sensitive to feature scale, StandardScaler normalization was applied before training. Predictions are the mean PM2.5 of the five nearest neighbors in feature space.

VI. RESULTS AND DISCUSSION

All three models were evaluated using Mean Absolute Error (MAE), Root Mean Squared Error (RMSE) and R-squared (R2). Lower MAE and RMSE indicate better prediction accuracy. Higher R2 indicates a better overall fit.

TABLE II. MODEL PERFORMANCE COMPARISON

Model	MAE	RMSE	R2 Score
Linear Regression	17.75	28.94	0.8222
Random Forest	9.24	19.27	0.921
KNN (k=5)	11.57	22.07	0.896

Random Forest Regression outperformed both Linear Regression and KNN across all three metrics. This is consistent with findings by Kumar and Pande who reported ensemble methods as superior to linear baselines on Indian air quality datasets. The advantage of Random Forest is its ability to model complex non-linear relationships between meteorological variables and PM2.5 that linear models cannot represent. Feature importance analysis confirmed that PM10 and CO were the most influential predictors. KNN was competitive but showed reduced accuracy in high-pollution outlier regions due to its local neighborhood sensitivity.

VII. CONCLUSION

This study presented a comprehensive three-year air quality analysis across four major Indian cities using CPCB monitoring data. Delhi was the most polluted city with PM2.5 levels nearly nine times the WHO safe limit [1], while Bengaluru maintained relatively better air quality. A consistent seasonal pollution cycle was observed — peaking in winter and declining during the monsoon — across all cities. Meteorological factors especially temperature and wind speed were found to influence PM2.5 significantly. Among the three models evaluated, Random Forest Regression achieved the best prediction accuracy, confirming the value of ensemble approaches [5] for non-linear air quality data. Future work could incorporate real-time data streams, multi-station spatial averaging per city and deep learning models for time-series aware prediction.

REFERENCES

[1] World Health Organization, "WHO Global Air Quality Guidelines: Particulate Matter (PM2.5 and PM10), Ozone, Nitrogen Dioxide, Sulfur Dioxide and Carbon Monoxide," WHO Press, Geneva, Switzerland, 2021.

[2] Central Pollution Control Board, "National Ambient Air Quality Standards," Ministry of Environment, Forest and Climate Change, Government of India, New Delhi, 2009. [Online]. Available: https://cpcb.nic.in

[3] K. Kumar and B. P. Pande, "Air pollution prediction with machine learning: a case study of Indian cities," International Journal of Environmental Science and Technology, vol. 20, pp. 5333-5348, 2023. doi: 10.1007/s13762-022-04241-5

[4] Doreswamy, H. K. S. Yogesh, K. M. Gad and I. Gad, "Forecasting air pollution particulate matter (PM2.5) using machine learning regression models," Procedia Computer Science, vol. 171, pp. 2057-2066, 2020. doi: 10.1016/j.procs.2020.04.221

[5] L. Breiman, "Random Forests," Machine Learning, vol. 45, no. 1, pp. 5-32, 2001. doi: 10.1023/A:1010933404324

[6] F. Pedregosa et al., "Scikit-learn: Machine Learning in Python," Journal of Machine Learning Research, vol. 12, pp. 2825-2830, 2011.